

Assignment 1: ETL with Python

Learning Objectives:

Implement an ETL pipeline in Python that ingests and integrates data from sources in different file formats, transforms the data, and loads the transformed data into a destination repository.

Due Date: January 19, 2025

Evaluation: 10% of the total grade

Instructions

Write Python code to perform the following tasks:

1. Create a suitable directory structure to host the data for each phase of the ETL process.
2. Download the zipped data folder from the URL and store in a temp location. Consider this a staging area / landing zone. The data consists of car prices in different file formats.
URL: <https://elasticbeanstalk-us-east-2-340729127361.s3.us-east-2.amazonaws.com/prices.zip>
3. Unzip the folder and copy data files under a different folder. Name the folder as you like.
4. Extract and integrate data from these files into a common Pandas DataFrame.
5. Perform a simple data transformation (double the price and round up to 2 decimal points), and load the transformed data in CSV format to a different folder (destination repository).
6. Log operations with appropriately tagged messages through out the ETL process in a logfile in a separate folder, e.g., Timestamp, [INFO] a message ..., Timestamp, [ERROR] another message ...

Submit the code preferably a Jupyter notebook in pdf format, along with the entire directory structure containing all files in appropriate sub-directories after execution.

Note: You may use a directory structure somewhat similar to as exemplified below:

Lab1 (as the root directory) with the following sub-directories:

- temp (staging area hosts the zip folder downloaded from the source URL),
- data (hosts the files extracted by unzipping the source),
- output (hosts the transformed data),
- logs (hosts the logfile)

Scoring Rubric

Category	Criteria	Scores
Code style	Code is well organized, clean and easy to follow with comments inserted in each cell	1.5
ETL functionality	All steps correctly performed and fully logged, results clearly displayed and observations briefly explained in the report	7.0
Lab Report	A well-structured brief report including learning objectives, brief explanation of results/observations, and concluding remarks.	1.5